

which are freely available on the Internet for download. The incoming CSV data format from EEPROM via COM port is stored in a string array. Simple software routines are included in the Delphi program to extract individual

weather data from the CSV format and displayed in Report List.

Reset button is used to reset the Delphi form. The user has to change the COM port number available in his PC and baud rate is fixed to 19200.

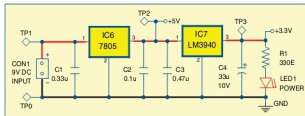


Fig. 4: Power supply

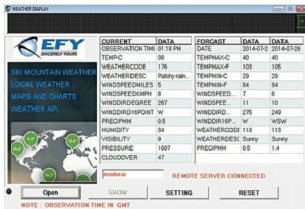


Fig. 5: Delphi front panel display

Construction and testing

An actual-size, double-side, solder-side PCB layout of the client-board is shown in Fig. 6, component-side track layout in Fig. 7 and the component layout in Fig. 8. An actual-size, single-side PCB layout for the power supply is shown in Fig. 9 and its component layout in Fig. 10.

Mount the components on the PCBs to save time and minimise assembly errors. Check proper connections before connecting the 5V and 3.3V

supplies to the PCB at CON4 and CON5, respectively.

If your PC does not have a serial COM port, you can connect it to USB port by using a USB-to-serial converter. In that case, first you need to install appropriate driver in your PC. After connecting the client board to your PC, launch the weatherdisplay.exe Delphi program. A window screen will open up similar to the one shown in Fig. 5 with blank data columns.

In the display panel of this window click on the Setting button and choose the COM port. Note that your PC COM port baud rate should be 19200. Enter the city name in the edit box, for instance, Madurai. Now switch on the client board. The 'TCP-Client' message should appear on the LCD. Vary VRI to adjust the contrast of the LCD to get clear visibility of the message.

Next connect one end of the Ethernet cable to the RJ45 connector in the client board and the other end to the Ethernet port of your broadband modem/router which has Internet connection. You need to configure your Internet connection in your PC from the control panel. You should disable wireless Internet connection. If Internet connection is successful, you will see the blinking of LED3 and LED4.

Now, click on Open button at the bottom of the screen. The IP address (192.168.0.102 in our case) should appear on the LCD. The Show button will be inactive first but after a few seconds it will get enabled and 'Remote server connected' message will appear at bottom of the screen. Upon clicking on the Show button weather data will get displayed in the report sheet as shown in Fig. 5.

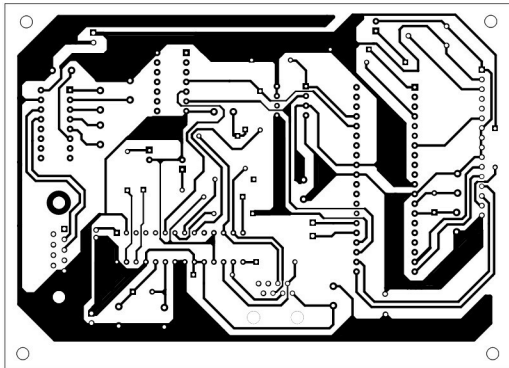


Fig. 6: An actual-size, double-side, solder-side PCB layout of the client board